FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Main)-2020

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. B

Sol. Let the length of rod is ℓ and co-ordinate of B is (x, y)

$\vec{v}_B = v_x\hat{i} + v_y\hat{j} = \sqrt{3}\hat{i} + v_y\hat{j}$

$x^2 + y^2 = \ell^2 \Rightarrow 2xv_x + 2yv_y = 0 \Rightarrow \sqrt{3} + \frac{y}{x}v_y = 0$

$\Rightarrow \sqrt{3} + \tan 60^o v_y = 0$ $v_y = -1$ m/s

$\vec{v}_B = \sqrt{3}\hat{i} - 1\hat{j}$

$|\vec{v}_B| = \sqrt{3+1} = 2m/s$

2. D

Sol. At focus path diff. = 0

$\Rightarrow I = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2 = 9I.$

3. C

Sol. For no deviation

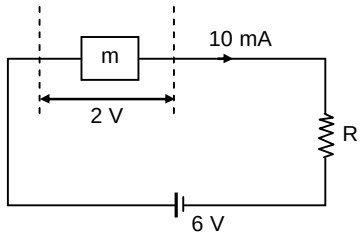
$(\mu_1 - 1)A_1 = (\mu_2 - 1)A_2$

$4^o(1.54 - 1) = (1.72 - 1)A_2$

$A_2 = \frac{4 \times 0.54}{0.72} = 3°.$

4. D

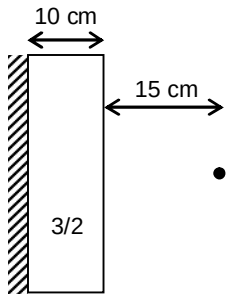
Sol. The potential drop across R is 4 V while current is 10 mA hence, $10 \times 10^{-3} \times R = 4$.
$R = 400\ \Omega$.



5. C

Sol. $X_{app} = \frac{10}{3/2} + \frac{10}{3/2} + \frac{15}{1}$

$= \frac{40}{3} + 15 = 28.33$



6. D

Sol. Apply Snell's law and use geometry.

7. A

Sol. $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$; $\frac{u}{v} = 1 + \frac{u}{f}$

$m = \frac{v}{u} = \frac{f}{u+f}$; $\frac{f}{-12+f} = -\left[\frac{f}{-16+f}\right]$ ($\because$ m is same in both the cases).

$f = 14$ cm.

8. B

Sol. If there is any charge inside the cavity of the conductor, the charge induced on the inner surface of the cavity is such that it will produce same amount of electric field but in opposite direction at each and every point outside the cavity as produced by the charge in the cavity.

9. A

Sol. $\Delta m = 0.1$ kg ; $\frac{\Delta m}{m} = \frac{0.1}{10}$

$\therefore$ Error = $\pm\frac{\Delta m}{m} \times 100 = \pm 1\%$.

10. B

Sol. Net force $F = m\, a_{net} = m\sqrt{a^2 + a^2} = ma\sqrt{2}$.

11. A

Sol. $\frac{hc}{\lambda} = RhCZ^2\left(1 - \frac{1}{4}\right)$

$\Rightarrow \frac{1}{\lambda_1} : \frac{1}{\lambda_2} : \frac{1}{\lambda_3} : \frac{1}{\lambda_4} = 1 : 1 : 4 : 9$

$\Rightarrow \lambda_1 : \lambda_2 : \lambda_3 : \lambda_4 = 1 : 1 : \frac{1}{4} : \frac{1}{9}$

12. C

Sol. $T = T_{shm} + 4\sqrt{\frac{2h}{g}}$

13. C

Sol. $f_1 = \frac{n}{2\ell}\sqrt{\frac{\gamma RT}{M}}$; $f_2 = \frac{n}{2(\ell + x)}\sqrt{\frac{\gamma RT}{M}}$

14. C

Sol. $F = mg$

$F = 2 \times 10^{-5} v = \frac{4}{3}\pi(1.5\times10^{-3})^3 \times 10^3 \times 10$

$\Rightarrow \quad v = \frac{2\pi}{3} \times (2.25) = 4.7$ m/s

15. C

Sol. From conservation of mechanical energy.

$\frac{1}{2}mv^2 = mgL + \frac{1}{2}m3(L - \ell)g$

$\therefore \quad v = \sqrt{g(5L - 3\ell)}$

16. B

Sol. $\tan\theta = \frac{X_c - X_L}{X_R}$

and $\cos\theta$ = power factor.

17. D

Sol. $W_{agent} - mgH = 0$

$W_{agent} = mgH$

18. B

Sol. $\because$ Field at the x-axis due to both current one same (along + z-axis) which is minimum distance between them.

19. D

Sol. $A_1V_1 = A_2V_2 \Rightarrow a\sqrt{2gh} = A\left(-\frac{dh}{dt}\right)$

$\Rightarrow \quad \frac{a}{A}\int_0^t dt = -A\int_{h_1}^{h_2} \frac{dh}{\sqrt{2gh}}$

20. B

Sol. If the capacitor has greater than E potential initially its potential will decrease and graph is first one. When capacitor has less than E potential initially it will follow the second graph.

SECTION – B

21. 3

Sol. $R = \frac{V^2}{gv_o} \cdot V = \frac{V^3}{gV_o}$; $R \alpha V^3$

22. 6

Sol. Time of height, $T = \frac{4V}{g\sqrt{7}}$, V = speed of projection

$V_H = \frac{V\sqrt{3}}{2\sqrt{7}}$

$V_\perp = \frac{3V}{2\sqrt{7}}$, when projectile with incline.

$\text{Tan}\ \alpha = \frac{1}{\sqrt{3}}$, $\alpha = 30°$.

SECTION – C

23. 00005.19

Sol. $t = 20\ \text{min} = \frac{1}{3}\text{hr}$

$1\ \text{km} = \sqrt{(6)^2 - V^2} \times \left(\frac{1}{3}\text{hr}\right)$

$V = 3\sqrt{3}\ \text{km/hr}$

24. 00000.99

Sol. Variation of moment of inertia with temperature

$I = I_o (1 + 2\alpha\ \Delta T)$

C.O.A.M:

$I_i\omega_i = I_f\ \omega_f$

25. 00002.32

Sol. $F_{\text{upthrust}} = F_{\text{surface tension}} + \text{Weight}$

$l^2 x\rho g = 4lT + mg$

$(.03)^2 \times 10^3 \times 10 = .009 + .2$

$9x = .209$

$x = .02322\ \text{m} = 2.322\ \text{cm}$

Chemistry

PART – II

SECTION – A

26. C
Sol. Because the bond angle is maximum (180°)

27. B
Sol. M_{eq} of HCl – M_{eq} of NaOH = 100 – 80 = 20

Moles of HCl in 900 mL = $\frac{20}{1000}$

Hence M = $\frac{1}{45}$

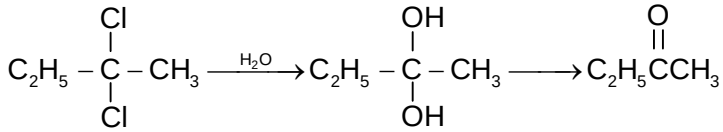
28. D
Sol. $\overline{E_K} = \frac{3}{2}nRT \Rightarrow \overline{E_K} \alpha T$

29. B
Sol. $K = \frac{\text{Rate}}{[X][Y]^{0.5}} = \frac{\text{mol}L^{-1}s^{-1}}{\text{mol}\,L^{-1} \times (\text{mol}\,L^{-1})^{0.5}} = \text{mol}^{-1/2}\,L^{1/2}s^{-1}$

30. A
Sol. $K_p = (p_B)^2 \Rightarrow p_B = \sqrt{K_p}$

31. B
Sol. Due to nearly same energy of the overlapping orbitals.

32. C
Sol. Geminal halides from ketone upon hydrolysis.



33. B
Sol. Due to completely filled 3d-orbtials.

34. B
Sol. It contains no symmetry element.

35. B
Sol. Fluorine cannot expand its valency from one.

36. B
Sol. Due to resonance or delocalization of unpaired electrons.

37. C
Sol. Both ethyl groups are attached to the same carbon atom.

38. D
Sol. $NaNO_3$ is not basic in nature. Hence it does not react with silica.

39. C
Sol. $B_3N_3H_6 + 3HCl \longrightarrow B_3N_3H_9Cl_3$

40. D
Sol. $Si(OH)_4 \xrightarrow{\Delta} SiO_2 + 2H_2O$

41. D
Sol. $Na_2B_4O_7 \xrightarrow{\Delta} 2NaBO_2 + B_2O_3$

42. B
Sol. CH_3 has no pi-electrons.

43. B
Sol. ClO_2 contains one unpaired electron.

44. B
Sol. Nucleophilic substitution takes place at carbonyl carbon. Cl^- is the best leaving group among the four.

45. B
Sol. The lone pair of N is in resonance with benzene ring. Hence, it can't accept proton from –COOH group.

SECTION – B

46. 6
Sol. Each O_2 molecule contains two unpaired electrons. Each O atom contains two unpaired electrons.

47. 5
Sol. Monobromination takes place at allylic positions. There are five allylic positions in the molecule.

SECTION – C

48. 00002.12
Sol. $\frac{d}{dr}\left(R^2 4\pi r^2\right) = 0$ at the maxima

On solving, r = $\frac{4a_0}{Z}$ at maxima

$= \frac{4 \times 0.53}{1} = 2.12\,\text{Å}$

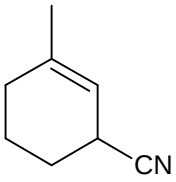
49. 18332.37

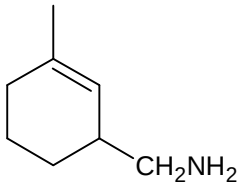
Sol. $\Delta H = nC_P\Delta T = 2\times\frac{7}{2}R\times(1315-2000)$

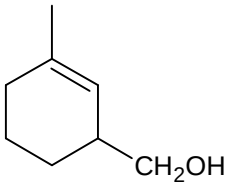
$= 2\times\frac{7}{2}\times 8.314\times 315 = 18332.37$ J

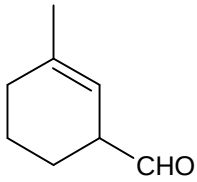
50. 00146.50

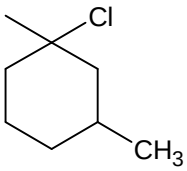
Sol.

P = 

Q = 

R = 

S = 

T = 

Mathematics

PART – III

SECTION – A

51. D

Sol. Let $y = f^{-1}(8x) \Rightarrow 8x = f(y) = 8y^3 + 3y = 8y^3\left(1+\frac{3}{8y^2}\right)$

$\Rightarrow 2x^{1/3} = 2y\left(1-\frac{3}{y^2}\right)^{1/3} \Rightarrow y = \frac{x^{1/3}}{\left(1+\frac{3}{y^2}\right)^{1/3}}$

Similarly, let $z = f^{-1}(x) \Rightarrow x = f(z) = 8z^3 + 3z = 8z^3\left(1+\frac{3}{8z^2}\right)$

$\Rightarrow x^{1/3} = 2z\left(1+\frac{3}{8z^2}\right)^{1/3} \Rightarrow z = \frac{x^{1/3}}{2\left(1+\frac{3}{8z^2}\right)^{1/3}}$

Thus, $\lim\limits_{x\to\infty} \frac{f^{-1}(8x) - f^{-1}(x)}{x^{1/3}}$

$= \lim\limits_{x\to\infty} \frac{y-z}{x^{1/3}}$

$= \lim\limits_{x\to\infty} \frac{y-z}{x^{1/3}}$

$= \lim\limits_{\substack{y\to\infty \\ z\to\infty}} \left(\frac{1}{\left(1+\frac{3}{8y^2}\right)^{1/3}} - \frac{1}{2\left(1+\frac{3}{8z^2}\right)^{1/3}}\right)$

$= 1-\frac{1}{2}$

$= \frac{1}{2}$

52. C

Sol. Let $I = \int_0^1 (x - f(x))^{2018}\, dx$

Put $x = f(t) \Rightarrow dx = f'(t)dt.$

Also $f(1) = f(f(0)) = 0$

$\therefore I = \int_1^0 (f(t) - t)^{2018} f'(t)dt$

$$= -\int_0^1 \left(t - f(t)\right)^{2018} f'(t)\,dt$$

$$= -\int_0^1 \left(t - f(t)\right)^{2018} \left(-1 + f'(t) + 1\right)dt$$

$$= -\int_0^1 \left(t - f(t)\right)^{2018} \left(-1 + f'(t) + 1\right)dt$$

$$= +\int_0^1 \left(t - f(t)\right)^{2018} \left(1 - f'(t)\right)dt - \int_0^1 \left(t - f(t)\right)^{2018} dt$$

$$= \int_0^1 \left(t - f(t)\right)^{2018} \left(1 - f'(t)\right)dt - I$$

$$2I = \int_0^1 \left(t - f(t)\right)^{2018} \left(1 - f'(t)\right)dt = \frac{\left(t - f(t)\right)^{2019}}{2019}\Bigg|_0^1$$

$$= \frac{1}{2019} + \frac{1}{2019} = \frac{2}{2019}$$

$$\Rightarrow I = \frac{1}{2019}$$

53. D

Sol. Let $P(h,k)$ be the point of intersection

Equation of tangent to $y^2 = 4ax$ is

$$y = m_1 x + \frac{a}{m_1} \qquad \text{.....(1)}$$

Also, equation of tangent to $x^2 = 4by$ is

$$y = m_2 x - bm_2^2 = -\frac{1}{m_1}x - \frac{b}{m_1^2} \quad \text{.....(2)} \qquad \left(\because m_1 m_2 = -1\right)$$

Since P (h, k) is their point of intersection

We have by substituting P (h, k) in (1) and (2)

$$hm_1^2 - km_1 + a = 0$$

$$km_1^2 + hm_1 + b = 0$$

Eliminating m_1

$$\frac{m_1^2}{-bk - ah} = \frac{m_1}{ak - bh} = \frac{1}{h^2 + k^2}$$

We get

$(h^2+k^2)(ah+bk)+(ak-bh)^2=0$

$\Rightarrow$ Locus of P (h, k) is

$(x^2+y^2)(ax+by)+(ay-bx)^2=0$

54. A

Sol. $\arg(z_2)+\arg(z_4)-\arg(z_1)-\arg(z_3)=\pi$

$\Rightarrow \arg\left(\frac{z_2z_4}{z_1z_3}\right)=\pi \Rightarrow \frac{z_2z_4}{z_1z_3}=-k$ (where $k>0$)

$\Rightarrow k=\left|\frac{z_2z_4}{z_1z_3}\right|=1$

$\Rightarrow z_2z_4=-z_1z_3$

$\left|z_1z_3-(z_1-z_3)(z_2-z_4)\right|^2-\left|(z_1-z_3)(z_2-z_4)\right|^2$

$=|z_1z_3|^2+\left|(z_1-z_3)(z_2-z_4)\right|^2-\bar{z}_1\bar{z}_3(z_1-z_3)(z_2-z_4)$

$-z_1z_3(\bar{z}_1-\bar{z}_3)(\bar{z}_2-\bar{z}_1)-\left|(z_1-z_3)(z_2-z_4)\right|^2$

$=1-\left\{\frac{(z_1-z_3)(z_2-z_4)}{z_1z_3}+\frac{(z_1-z_3)(z_2-z_4)}{z_2z_4}\right\}=1$

55. B

Sol. $\begin{vmatrix}1&1&1\\1&1&1\\0&0&0\end{vmatrix}=0$

$\begin{vmatrix}1&0&0\\0&1&0\\0&0&1\end{vmatrix}=1,-1$ can be obtained by interchanging any two adjacent row/column

$\begin{vmatrix}1&1&0\\0&1&1\\1&0&1\end{vmatrix}=2,-2$ may be obtained by interchanging any two adjacent row/column

Let $M=\begin{bmatrix}a_1&b_1&c_1\\a_2&b_2&c_2\\a_3&b_3&c_3\end{bmatrix}$ be matrix with only 0 or 1 as its entries.

Now, every 2×2 sub – matrix of M has determinant 0 or ± 1

So if some row has at least one '0' , then expanding the determinant w.r.t. that row we see that it can be at most 2.

Hence for $\det(M)$ to be equal to 3, all rows must have only 1's, in which case $\det(M) = 0$, So, there is no matrix M of the given type whose determinant is 3.

56. D

Sol. _R_R_R_R_R_R_
1 2 3 4 5 6

Number of ways of select 4 places for the four green balls from available 6 spaces $= {}^6C_4 = 15$ ways

But we have to exclude those arrangements in which three (or more) red balls are together.

Gaps 2, 3 cannot be left vacant.

Similarly 3, 4 cannot be left vacant

And 4, 5 cannot be left vacant

$\Rightarrow$ Required number of ways = 15 – 3 = 12

57. B

Sol. $$\lim_{x\to 0}\frac{f(x)}{x^k} = \lim_{x\to 0}\frac{\frac{1}{2}x\sin x - 1 + \cos x}{x^k}$$

$$= \lim_{x\to 0}\frac{\frac{1}{2}\sin x + \frac{1}{2}x\cos x - \sin x}{k\,x^{k-1}}$$

$$= \lim_{x\to 0}\frac{\frac{1}{2}\cos x + \frac{1}{2}\cos x - \frac{1}{2}x\sin x - \cos x}{k(k-1)x^{k-2}}$$

$$= \lim_{x\to 0}\frac{-\frac{1}{2}\sin x}{k(k-1)x^{k-3}}$$

If $k-3=1$ i.e. $k=4$, $\lim_{x\to 0}\frac{f(x)}{x^k} = \frac{-\frac{1}{2}}{4\times 3} = \frac{-1}{24}$

58. D

Sol. Obviously $x\tan 10^\circ + y\tan 20^\circ + \tan 30^\circ = 0$ and $x\tan 120^\circ + y\tan 220^\circ + \tan 320^\circ = 0$ are asymptotes of a hyperbola.

Since, these lines are perpendicular, we have a rectangular hyperbola

$\therefore e = \sqrt{2}$

59. D

Sol. $\ell n f(x) = \lim\limits_{n\to\infty} n\ell n\left(\cos\left(\frac{1}{n^x}\right)\right)$

Let $n = \frac{1}{\mu}$

$\Rightarrow \ell n f(x) = \lim\limits_{\mu\to 0^+} \frac{\ell n\left(\cos\mu^x\right)}{\mu} = \lim\limits_{x\to 0^+} -\tan\left(\mu^x\right).x.\mu^{x-1}$

$= -\lim\limits_{\mu\to 0^+} \frac{\tan\left(\mu^x\right)}{\mu^x}.x.\mu^{2x-1}$

$$= \begin{cases} -\infty, & 0 < x < \frac{1}{2} \\ -\frac{1}{2} & x = \frac{1}{2} \\ 0 & x > \frac{1}{2} \end{cases}$$

$$f(x) = \begin{cases} 0, & 0 < x < \frac{1}{2} \\ e^{-\frac{1}{2}} & x = \frac{1}{2} \\ 1 & x > \frac{1}{2} \end{cases}$$

So, clearly $c = \frac{1}{2}$ and $f(c) = e^{-1/2}$

60. B

Sol. $x^5 + x + 1 \equiv \left(x^2 + x + 1\right)\left(x^3 - x^2 + 1\right)$

Thus $3x^4 + 2x^3 - 2x + 1 \equiv x^3 - x^2 + 1 + \left(3x^2 - 2x\right)\left(x^2 + x + 1\right)$

61. C

Sol. $\frac{1}{n}\sum\limits_{0\le i<j\le n}\sum \frac{1}{\sqrt{(n+i)(n+j)}}$

$= \frac{1}{2n}\left(\sum\limits_{i=1}^{n}\frac{1}{\sqrt{n+i}}\right)^2 - \frac{1}{2n}\sum\limits_{i=1}^{n}\frac{1}{n+i}$

$$= \frac{1}{2}\left(\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{1}{\sqrt{n+i}}\right)^2 - \frac{1}{2n}\sum_{i=1}^{n}\frac{1}{n+i}$$

$$= \frac{1}{2}\left(\frac{1}{n}\sum_{i=1}^{n}\frac{1}{\sqrt{1+\frac{i}{n}}}\right)^2 - \frac{1}{2n}\sum_{i=1}^{n}\frac{1}{n+i}$$

$$= \frac{1}{2}\left(\int_0^1 \frac{1}{\sqrt{1+x}}dx\right)^2 - \frac{1}{2n}\sum_{i=1}^{n}\frac{1}{n+i}$$

$$= \frac{1}{2}\left(2\sqrt{2}-2\right)^2 - L$$

$$= 2\left(3-2\sqrt{2}\right) - L$$

$$L = \lim_{n\to\infty}\frac{1}{2n}\sum_{i=1}^{n}\frac{1}{n+i}$$

Now, $0 \le \frac{1}{n}\sum_{i=1}^{n}\frac{1}{n+1} \le \frac{1}{n+1}$

$\Rightarrow L = 0$

So, required limit $= 2\left(3-2\sqrt{2}\right)$

62. D

Sol. $S_n = \sum_{j\ge 0}(-1)^j\ {}^{n-j}C_j$

$$S_{n+1} = \sum_{j\ge 0}(-1)^j\ {}^{n+1-j}C_j$$

$$= \sum_{j\ge 0}(-1)^j\left({}^{n-j}C_j + {}^{n-j}C_{j-1}\right)$$

$$= \sum_{j\ge 0}(-1)^j\left({}^{n-j}C_j\right) - \sum_{j\ge 0}(-1)^{j-1}\left({}^{n-1-(j-1)}C_{j-1}\right)$$

$= S_n - S_{n-1}$

$S_1 = 1; S_2 = 0,$ so S_n is $1,0,-1,-1,0,1,1,0,-1,-1,0,1,.....$

Which is a periodic sequence.

63. C

Sol. ${}^{36}C_3 - 36 - 36(36-4) - (18\times 30) = 5412$

64. B

Sol. $f(-1) = 1+2a-3+a = 3a-2$

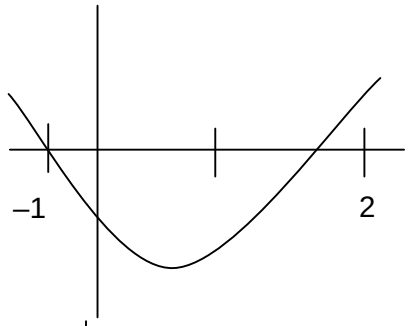
$f(2) = 4+6-4a+a = 10-3a$

Case I

$f(-1)f(2) < 0 \Rightarrow (3a-2)(10-3a) < 0$

$\Rightarrow a \in \left(-\infty, \frac{2}{3}\right) \cup \left(\frac{10}{3}, \infty\right)$

Case II



$f(-1) = 0 \Rightarrow a = \frac{2}{3}$

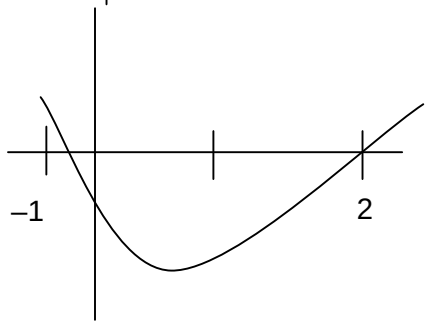
$f(2) = 10 - 3a = 10 - 2 = 8 > 0$

Also, x – coordinate of vertex lies in (–1, 2)

$\Rightarrow$ Exactly one root lies between –1 and 2

for this value of a $\left(= \frac{2}{3}\right)$

Case III



$f(2) = 0 \Rightarrow a = \frac{10}{3}$

$f(-1) = 3a - 2 = 3\left(\frac{10}{3}\right) - 2 = 8 > 0$

x – coordinate of vertex $= \frac{2a-3}{2} = \frac{11}{6} < 2$

$\Rightarrow$ Exactly one root lies between –1 and 2

for this value of a $\left(= \frac{10}{3}\right)$

$\therefore a \in \left(-\infty, \frac{2}{3}\right] \cup \left[\frac{10}{3}, \infty\right)$

65. C

Sol. $2a + (m-1)d = \frac{2n}{m}$; $2a + (n-1)d = \frac{2m}{n} \Rightarrow d = \frac{-2(m+n)}{mn}$

and $2a + \frac{(m+n-2)d}{2} = \frac{n}{m} + \frac{m}{n}$ (on adding)

$\Rightarrow 2a + (m+n-1)d + \frac{(m+n-2)d}{2} - (m+n-1)d = \frac{n}{m} + \frac{m}{n}$

$\Rightarrow 2a + (m+n-1)d - \frac{(m+n)}{2}d = \frac{m^2+n^2}{mn}$

$2a + (m+n-1)d = \frac{m^2+n^2}{mn} + \frac{(m+n)d}{2}$

$= \frac{m^2+n^2}{mn} + \frac{m+n}{2}\left\{\frac{-2(m+n)}{mn}\right\}$

$= \frac{m^2+n^2-(m+n)^2}{mn} = -2$

Required sum $= \frac{m+n}{2}\left[2a + (m+n-1)d\right] = -(m+n)$

Thus C.

66. B

Sol. Uncle Sam will not be able to park when none of the four left over parking spaces are adjacent.

Number of ways in which the 12 drivers may leave 4 spaces $= {}^{16}C_4$

Number of ways in which the 12 – drivers leave 4 spaces so that none of four left over spaces are adjacent $= {}^{13}C_4$

$\Rightarrow$ Required probability = 1 – P (Uncle Sam doesn' t park)

$$= 1 - \frac{{}^{13}C_4}{{}^{16}C_4} = 1 - \frac{13.12.11.10}{16.15.14.15} = 1 - \frac{12.11.10}{16.15.14} = 1 - \frac{11}{28} = \frac{17}{28}$$

67. D

Sol. Applying LMVT on $[0, 4]$, we get $f(c_1) = \frac{f(4) - f(0)}{4 - 0} = \frac{1 - 0}{4} = \frac{1}{4}$ for atleast one $c_1 \in (0,4)$. Hence (P).

Applying Rolle's theorem on $[4,8]$, we get $f'(d) = 0$ for some $d \in (4, 8)$

Since derivative is continuous, by Intermediate Value Properly Theorem for function $y = f'(x)$ on the interval $[c_1, d]$ there exists at least one $c \in [c_1, d]$ where $f'(c) = \frac{1}{2}$

$\left(\because 0 < \frac{1}{12} < \frac{1}{4}\right)$

Hence, (Q).

By LMVT on $[0, 4]$ we have $f'(c_1) = \frac{1}{4}$ and by IVPT on $[0,4]$ we have $f(c_2) = \frac{1}{2}$ for some $c_2 \in (0, 4)$

$\Rightarrow 8f'(c_1) f(c_2) = 1$

Hence (R)

$$f(f(8)) - f(f(0)) = \frac{f(f(8)) - f(f(0))}{f(8) - f(0)} \frac{f(8) - f(0)}{8 - 0} \times 8$$

$$= 8f'(e_1) f'(e_2)$$

$$= 8(f'(c))^2 \quad (f'(c) \text{is G.M. of } f'(e_1) \text{and } f'(e_2))$$

Hence (S)

68. A

Sol. Point of intersections are $(1,0,0), \left(\frac{1}{2}, \frac{1}{2}, 0\right), \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$

69. D

Sol. Since ellipse touches X – axis at O(0, 0), Y – axis is therefore normal to be ellipse.

$\Rightarrow$ Y – axis is thus internal angular bisector of $\angle F_1OF_2$ (F_1 and F_2 being the two focii).

$\Rightarrow$ Equation of OF_1 and OF_2 can therefore be assumed as $y = kx$ and $y = -kx$.

Equation of major axis is $y = x + 1$. Solving OF_1 and OF_2 with major axis.

We get $F_1 \equiv \left(\frac{1}{k-1}, \frac{k}{k-1}\right)$ and $F_2 \equiv \left(\frac{1}{k+1}, -\frac{k}{k+1}\right)$

$C \equiv$ Mid point of $F_1F_2 \equiv (3, 4)$

$\Rightarrow k = \frac{2}{\sqrt{3}} \Rightarrow F_1 \equiv \left(3+2\sqrt{3}, 4+2\sqrt{3}\right)$ and $F_2 = \left(3-2\sqrt{3}, 4-2\sqrt{3}\right)$

$CF_1 = ae = \sqrt{\left(2\sqrt{3}\right)^2 + \left(2\sqrt{3}\right)^2} = 2\sqrt{6}$

Also major axis

$= 2a = OF_1 + OF_2 = \sqrt{\left(3+2\sqrt{3}\right)^2 + \left(4+2\sqrt{3}\right)^2} + \sqrt{\left(3-2\sqrt{3}\right)^2 + \left(4-2\sqrt{3}\right)^2} = 4\sqrt{7}$

$\Rightarrow a = 2\sqrt{7}$

$\Rightarrow e = \sqrt{\frac{6}{7}}$

70. B

Sol. Let $S : (\sim p \wedge q) \vee (p \wedge \sim q)$

$\Rightarrow \sim S : \sim \left[(\sim p \wedge q) \vee (p \wedge \sim q)\right]$

$\Rightarrow \sim S : \sim (\sim p \wedge q) \wedge \sim (p \wedge \sim q)$

$\Rightarrow \sim S : (p \vee \sim q) \wedge (\sim p \vee q)$

SECTION – B

71. 2

Sol. $f'(x) = 3x^2 + 2x - a\sin x$

Clearly $f'(0) = 0$

$\Rightarrow x = 0$ must be point of minima of $y = f'(x)$ (if sign of derivative should not change)

i.e. $f''(0) = 0$

$\Rightarrow f''(x) = 6x + 2 - a\cos x$

$f''(0) = 2 - a = 0 \Rightarrow a = 2$

$f'''(x) = 6 + a\sin x \Rightarrow f'''(0) = 6 + a = 6 + 2 = 8 > 0$

72. 2

Sol. Numerators of the terms are 1, 1, 2, 3, 5, 8, 13,…… which are terms of Fibbonacci sequence

$F_1 = 1; F_2 = 1; F_3 = 2; .....; F_n = F_{n-1} + F_{n-2}$

$S = \sum_{n=1}^{\infty} \frac{F_n}{2^n} = \frac{1}{2} + \frac{1}{4} + \sum_{n=3}^{\infty} \frac{F_{n-1} + F_{n-2}}{2^n}$

$\Rightarrow S = \frac{1}{2} + \frac{1}{4} + \sum_{n=3}^{\infty} \frac{F_{n-1}}{2^n} + \sum_{n=3}^{\infty} \frac{F_{n-2}}{2^n}$

$\Rightarrow S = \frac{1}{2} + \frac{1}{4} + \frac{1}{2}\sum_{n=3}^{\infty} \frac{F_{n-1}}{2^{n-1}} + \frac{1}{4}\sum_{n=3}^{\infty} \frac{F_{n-2}}{2^{n-2}}$

$$\Rightarrow S = \frac{1}{2} + \frac{1}{4} + \frac{1}{2}\left(S - \frac{1}{2}\right) + \frac{1}{4}S$$

$$\Rightarrow S = 2$$

SECTION – C

73. 00065.00

Sol. $(x+5)^2 = (x+1)^2 + 36$

$10x + 25 = 2x + 1 + 36$

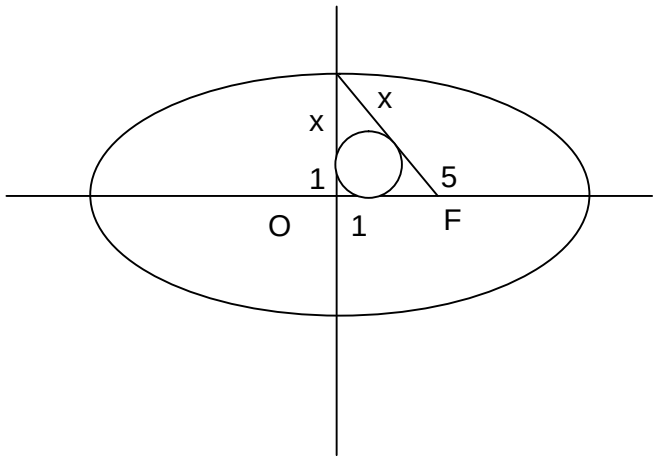
$8k = 37 - 25 = 12$

$x = \frac{12}{8} = \frac{3}{2}$

$CD = 2(x+1) - \frac{5}{2} \times 2 = 5$

$AB = 2(x+5) = \frac{13}{2} \times 2 = 13$

(AB) (CD) = 65



74. 00003.00

Sol. $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9,$

We get $\vec{a}.\vec{b} + \vec{b} + \vec{c} + \vec{c}.\vec{a} = -\frac{3}{2} \Rightarrow 2\vec{a}.\vec{b} + 2\vec{b}.\vec{c} + 2\vec{c}.\vec{a} + 3 = 0$

$\Rightarrow |\vec{a} + \vec{b} + \vec{c}|^2 = 0$

$\Rightarrow \vec{a} + \vec{b} + \vec{c} = 0$

So, $\vec{a}, \vec{b}, \vec{c}$ from an equilateral triangle

$\Rightarrow \vec{a}.\vec{b} = \vec{b}.\vec{c} = \vec{c}.\vec{a} = -\frac{1}{2}$

$\Rightarrow |2\vec{a} + 5\vec{b} + 5\vec{c}|^2 = 4a^2 + 25b^2 + 25c^2 + 20\vec{a}.\vec{b} + 50\vec{b}.\vec{c} + 20\vec{c}.\vec{a}$

$= 54 + 90\left(-\frac{1}{2}\right)$

$= 54 - 45$

= 9

$\Rightarrow |2\vec{a} + 5\vec{b} + 5\vec{c}| = 3$

75. 00002.00

Sol. $\sum x_i - 45 = 9 \Rightarrow \sum x_i = 54$

Now, $\sum x_i^2 - 10 \times 54 + 25 \times 9 = 45 \Rightarrow \sum x_i^2 = 360$

$\therefore \quad \sigma = \sqrt{\frac{360}{9} - \left(\frac{54}{9}\right)^2} = 2$